

# SQL Research Assignment

## Section A: Data Fundamentals

### 1. What is a Database?

A database is an organised collection of information that can be stored, managed, searched and updated using a computer system. Database are important because they allow businesses and organisations to store large amounts of information efficiently and retrieve it when needed.

- Banks use databases to store customers accounts transactions and payment information.
- Hospitals use databases to store patient records appointments and medical information.

### 2. What is a data warehouse?

A data warehouse is an enterprise system used for the analysis and reporting of structured and semi-structured data from multiple sources, such as point-of-sale transactions marketing automation customer relationship management.

A company may use a data warehouse to combine sales, customers and financial information so managers can identify trends and make better business decisions.

### 3. What is a data lake?

A centralised, highly scalable repository designed to store, process, and analyse massive amounts of structured, semi-structured, and unstructured data in its native format.

A data warehouse usually stores process and organised data, while a data lake can store raw data before it has been processed. A business may prefer a data lake when it needs to store large amounts of different types of data for future analysis, machine learning or experimentation.

### 4. What is a Data Mart?

A data mart is a smaller section of a data warehouse that focuses on a particular department or business area. A company could have a sales data mart containing only sales-related information.

### 5. Structured, Semi-Structured and Unstructured Data

- **Structured data:** Data organised into a fixed format, such as rows and columns in a database. eg. a customer table containing names, addresses and phone numbers.
- **Semi-structured data:** Data that does not follow a strict table structure but contains labels or other organisational information.

Unstructured data: Data without a predestined structure: example photographs, videos etc

## 6. What is Data Modeling

Data modeling is the process of designing how data will be organised, stored and related to other data. It is important because a good data model helps prevent duplication, improves accuracy and makes information easier to retrieve.

Two types:

- Relational data model: Organises information into tables containing rows and columns
- Document data model: Stores information as documents, often using formats such as JSON

## 7. What is Data Mining

Data mining is the process of examining large amounts of data to discover useful patterns, relationships and trends.

Businesses use data mining to understand customers and improve decision-making

eg.

1. An online shop analysing customer purchases to recommend products.
2. A bank analysing transactions to identify unusual activity that could indicate fraud

## 8. What is a DBMS?

A ~~data~~ database Management System (DBMS) is software that allows users to create, store, manage and retrieve information from database.

eg.

MySQL: A relational database commonly used for websites and business applications.

PostgreSQL: An advanced relational database suited for complex applications and data analysis.

Oracle Database: A powerful often used by large organisations and enterprises.

MongoDB: A document-based database that stores data in flexible documents and is useful for applications where data structures may change frequently.

## 9. What is ETL?

ETL stands for Extract, Transform and Load.

- Extract: Collect data from different sources
- Transform: Clean, organise and convert the data into a useful format
- Load: Put the transformed data into a database, data warehouse or other storage system.

ETL is important because data often comes from different systems and needs to be cleaned and combined before it can be analysed.

## 10. Common data storage formats

- CSV: Stores data in rows and columns separated by commas. It is used for simple tabular data and spreadsheets.
- JSON: Stores data using a structure of objects and values. It is commonly used for APIs and web applications.

Parquet: A column-based format designed for efficient storage and analysis of large datasets.

- AVRO: A data serialisation format often used in large-scale systems and data pipelines.

CSV is useful for simple datasets, JSON is useful for applications, and Parquet and AVRO are often more suitable for large scale data processing.

## SECTION B - ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

### 11. What is Artificial Intelligence

Artificial Intelligence (AI) is technology that enables computers to perform tasks that normally require human intelligence, such as understanding language, recognising images, making predictions and solving problems.

Narrow AI is designed to perform specific tasks eg. a spam filter can identify unwanted email.

General AI : refers to a theoretical form of AI that could perform a wide range of ~~intelligence~~ intellectual tasks at a human-like level. A fully realised General AI does not currently exist.

2. What is Machine learning?

Machine Learning (ML) is a branch of AI where computers learn patterns from data and use those patterns to make predictions or ~~data~~ decision.

Traditionally programming usually involves giving a computer specific rules and instructions. In machine learning the computer learns pattern from examples.

3 types:

Supervised learning: The model learns from labelled examples. Eg. predicting house prices from previous house sales.

Unsupervised: The models find patterns in data without labelled answers. Example: grouping customers according to their behaviour.

Reinforcement learning: A system learns through rewards and penalties as it interacts with an environment. Eg training an AI system to play a game.

### 13. What is Deep learning

Deep learning is a type of machine learning that uses neural networks with many layers to learn complex patterns.

The relationship is:

Artificial Intelligence  $\rightarrow$  Machine learning  $\rightarrow$  Deep Learning

A neural network is a computer system inspired loosely by the way biological brains process information. It contains connected nodes called neurons. The network processes input data through different layers and adjusts its internal parameters during training to improve its results.

### 14. What is NLP?

Natural Language Processing (NLP) is a field of AI that enables computers to understand, process and generate human language.

Three everyday examples:

- Voice assistants such as Siri or Google Assistant
- Automatic translation tools
- Email spam detection and prediction.

15. What is an LLM?

A Large Language Model (LLM) is an AI model trained on very large amounts of ~~text~~ data so that it can understand and generate human-like language.

Unlike many traditional machine-learning models that are designed for a specific task,

LLMs can perform many language-related tasks such as answering questions, summarizing information, translating and generating text.

2 examples

Chat GPT and Claude. They can be used for tasks such as writing, summarization, coding, answering questions

16. What is Generative AI?

Generative AI is AI that can create new content based on patterns learned from training data.

It can generate text, images, audio, video and computer code.

At a high level, a generative AI model needs an instruction or prompt and generates an output based on patterns it learned during training.

Eg.

Chat-GPT: Generator and works with text, and can also perform other AI-assisted tasks

Claude: An AI assistant that can generate and analyse text

DALL-E: Generates images from text descriptions

## 17. What is a training Dataset

A training dataset is a collection of examples used to teach an AI or machine-learning model how to perform a task.

The quality of training data is important because the model learns patterns from that data.

If the data is inaccurate, incomplete or biased, the resulting model may produce inaccurate or unfair results

For example, if a facial-recognition system is trained using data that poorly represents certain groups of people, it may perform less accurately for those groups

## 18. What is Computer vision

Computer vision is a field of AI that allows computers to analyse and interpret images and videos

Machines can process images as numerical values and use trained models to identify patterns such as objects, faces, text or shapes.

Two examples:

- facial ~~express~~ recognition and security systems
- self driving or driver assistance systems that identify vehicles, pedestrians and road signs

## Section C - APIs, Tools and Developer Concepts

### 19. What is an API

An API or application Programming Interface, is a way for different software applications to communicate with each other.

A simple analogy is a restaurant. The customer is the application making a request, the waiter is the API, and the kitchen is the system providing the requested information. The waiter ~~takes~~ takes the request to the kitchen and brings the result back.

### 20. What is an API key?

An API key is a unique piece of information used to identify or authentication an application when it accesses an API.

API keys should be kept secret and stored securely, such as in environment variables or secure secret-management systems. They should not be published in source code or public website.

## 21. What is a Rest API

Rest stands for Representational State Transfer. A Rest API is an API design approach that commonly uses HTTP to allow applications to communicate with resources.

Common HTTP include:

- GET: Retrieve information
- POST: Create or submit new information
- PUT: Update or replace existing information
- DELETE: Remove information

## 22. What is JSON

JSON stands for Java Script Object Notation. It is a lightweight format used to store and exchange structure information.

It is commonly used with API because it is easy for both humans and computers to read and process.

JSON

```
{  
  "name": "John",  
  "age": 25,  
  "gender": "male"
```

## 23. What is Prompt Engineering?

Prompt engineering is the process of designing clear and effective instructions for an AI system.

It is important because the quality and clarity of a prompt can affect the usefulness of the AI's response.

Poor prompt

"Tell me about data"

Improved prompt

"Explain what data analytics is to a beginner in simple language and give an example of how business use it"

Poor prompt

"Write an email"

Improved prompt

"Write a professional but friendly email to a customer explaining that their payment has been received and their order is now being processed"

## 24. What is a MAP Function

A Map function applies a specific operation to each item in a collection of data and produces a new collection of results.